Unit 5 Python Programs

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Creating a common dataset
data = {
    "Name": ["Arun", "Bala", "Charan", "Divya", "Esha", "Farah", "Arun", "Hari"],
    "Age": [21, 22, np.nan, 24, 23, 22, 21, np.nan],
    "Department": ["ECE", "CSE", "ECE", "IT", "CSE", "ECE", "ECE", "IT"],
    "Marks": [85, 92, 78, np.nan, 88, 95, 85, 70],
    "Attendance": [90, 85, 78, 92, np.nan, 88, 90, 75]
}

df = pd.DataFrame(data)

print(df)
```

```
     Name   Age Department  Marks  Attendance
0    Arun  21.0        ECE   85.0        90.0
1    Bala  22.0        CSE   92.0        85.0
2  Charan   NaN        ECE   78.0        78.0
3   Divya  24.0         IT    NaN        92.0
4    Esha  23.0        CSE   88.0         NaN
5   Farah  22.0        ECE   95.0        88.0
6    Arun  21.0        ECE   85.0        90.0
7    Hari   NaN         IT   70.0        75.0
```

1. Filtering data

```
filtered_data = df[df["Marks"] > 85]

print(filtered_data)
```

```
    Name   Age Department  Marks  Attendance
1   Bala  22.0        CSE   92.0        85.0
4   Esha  23.0        CSE   88.0         NaN
5  Farah  22.0        ECE   95.0        88.0
```

2. Filtering Data with Multiple Conditions

```
filtered_data = df[
    (df["Marks"] > 80) &
    (df["Attendance"] > 85)
]

print(filtered_data)
```

```
    Name   Age Department  Marks  Attendance
0   Arun  21.0        ECE   85.0        90.0
5  Farah  22.0        ECE   95.0        88.0
6   Arun  21.0        ECE   85.0        90.0
```

3. Sorting Data in Both Orders

Ascending Order

```
ascending_data = df.sort_values(by="Marks", ascending=True)

print(ascending_data)
```

```
     Name   Age Department  Marks  Attendance
7    Hari   NaN         IT   70.0        75.0
2  Charan   NaN        ECE   78.0        78.0
6    Arun  21.0        ECE   85.0        90.0
0    Arun  21.0        ECE   85.0        90.0
4    Esha  23.0        CSE   88.0         NaN
1    Bala  22.0        CSE   92.0        85.0
5   Farah  22.0        ECE   95.0        88.0
3   Divya  24.0         IT    NaN        92.0
```

Descending Order

```
descending_data = df.sort_values(by="Marks", ascending=False)

print(descending_data)
```

```
     Name   Age Department  Marks  Attendance
5   Farah  22.0        ECE   95.0        88.0
1    Bala  22.0        CSE   92.0        85.0
4    Esha  23.0        CSE   88.0         NaN
6    Arun  21.0        ECE   85.0        90.0
0    Arun  21.0        ECE   85.0        90.0
2  Charan   NaN        ECE   78.0        78.0
7    Hari   NaN         IT   70.0        75.0
3   Divya  24.0         IT    NaN        92.0
```

4. Handling Missing Values

Detecting Missing Values

```
missing_values = df.isnull()

print(missing_values)
```

```
    Name    Age  Department  Marks  Attendance
0  False  False       False  False       False
1  False  False       False  False       False
2  False   True       False  False       False
3  False  False       False   True       False
4  False  False       False  False        True
5  False  False       False  False       False
6  False  False       False  False       False
7  False   True       False  False       False
```

Counting Missing Values

```
missing_count = df.isnull().sum()

print(missing_count)
```

```
Name          0
Age           2
Department    0
Marks         1
Attendance    1
dtype: int64
```

5. Removing Missing Values

```
clean_df = df.dropna()

print(clean_df)
```

```
    Name   Age Department  Marks  Attendance
0   Arun  21.0        ECE   85.0        90.0
1   Bala  22.0        CSE   92.0        85.0
5  Farah  22.0        ECE   95.0        88.0
6   Arun  21.0        ECE   85.0        90.0
```

6. Removing Duplicate Data

```
unique_df = df.drop_duplicates()

print(unique_df)
```

```
     Name   Age Department  Marks  Attendance
0    Arun  21.0        ECE   85.0        90.0
1    Bala  22.0        CSE   92.0        85.0
2  Charan   NaN        ECE   78.0        78.0
3   Divya  24.0         IT    NaN        92.0
4    Esha  23.0        CSE   88.0         NaN
5   Farah  22.0        ECE   95.0        88.0
7    Hari   NaN         IT   70.0        75.0
```

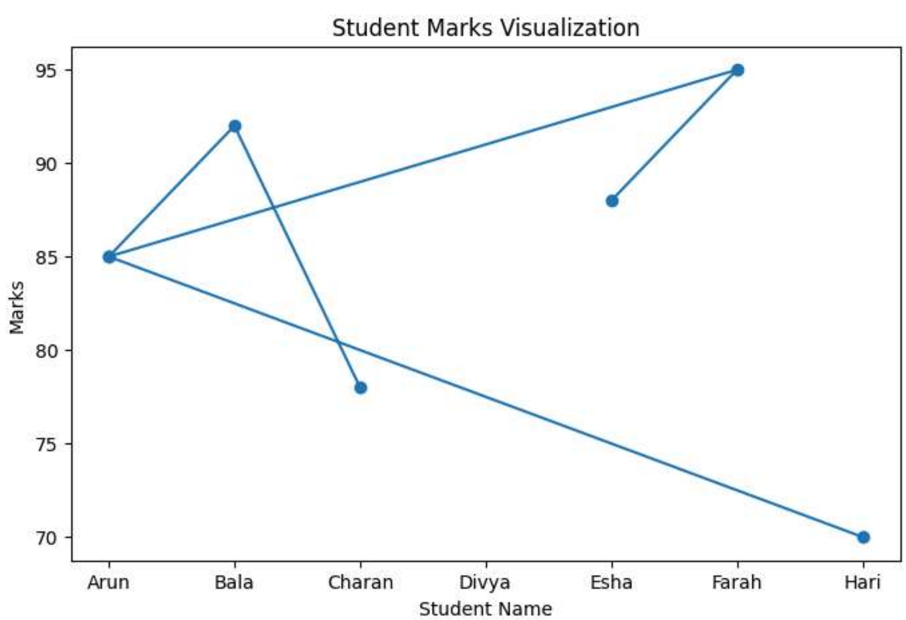
7. Aggregating the Data

```
aggregation = df.groupby("Department").agg({
    "Marks": "mean",
    "Attendance": "mean",
    "Age": "mean"
})

print(aggregation)
```

```
            Marks  Attendance        Age
Department
CSE         90.00        85.0  22.500000
ECE         85.75        86.5  21.333333
IT          70.00        83.5  24.000000
```

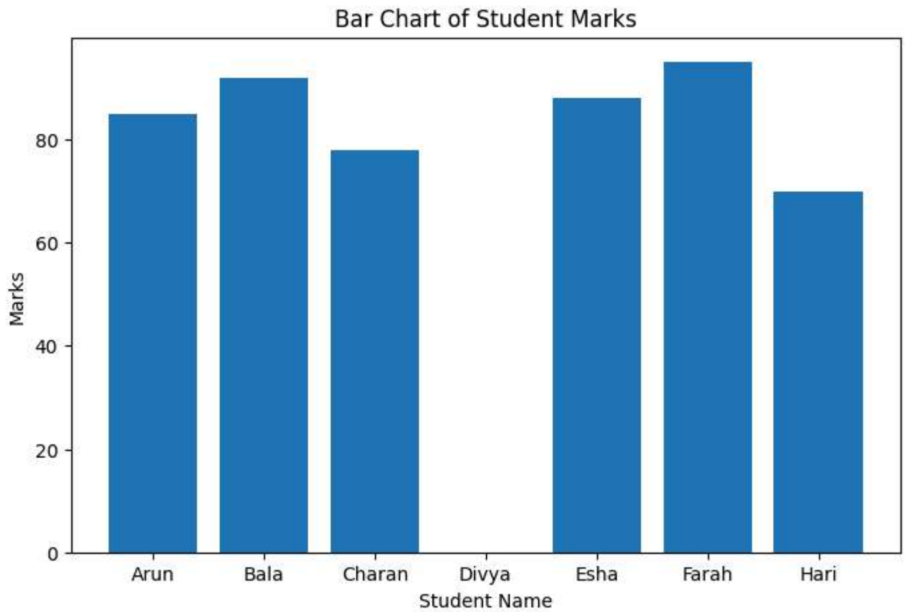
8. Data Visualization (Line Plot)

```
plt.figure(figsize=(8, 5))

plt.plot(df["Name"], df["Marks"], marker="o")

plt.xlabel("Student Name")
plt.ylabel("Marks")
plt.title("Student Marks Visualization")

plt.show()
```



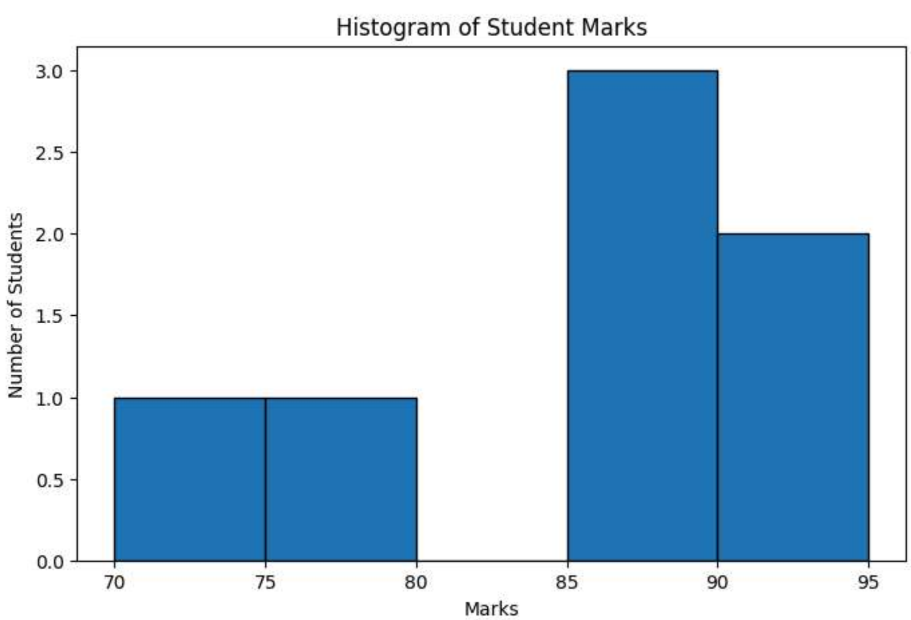
9. Bar Chart

```
plt.figure(figsize=(8, 5))

plt.bar(
    df["Name"],
    df["Marks"]
)

plt.xlabel("Student Name")
plt.ylabel("Marks")
plt.title("Bar Chart of Student Marks")

plt.show()
```



10. Histogram

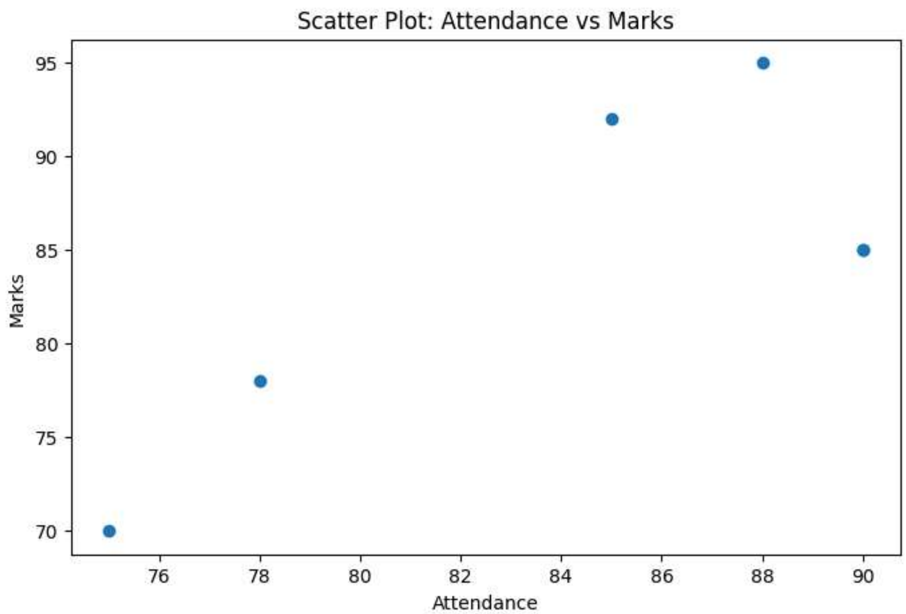
```
plt.figure(figsize=(8, 5))

plt.hist(
    df["Marks"].dropna(),
    bins=5,
    edgecolor="black"
)

plt.xlabel("Marks")
plt.ylabel("Number of Students")
plt.title("Histogram of Student Marks")

plt.show()
```



11. Scatter Plot

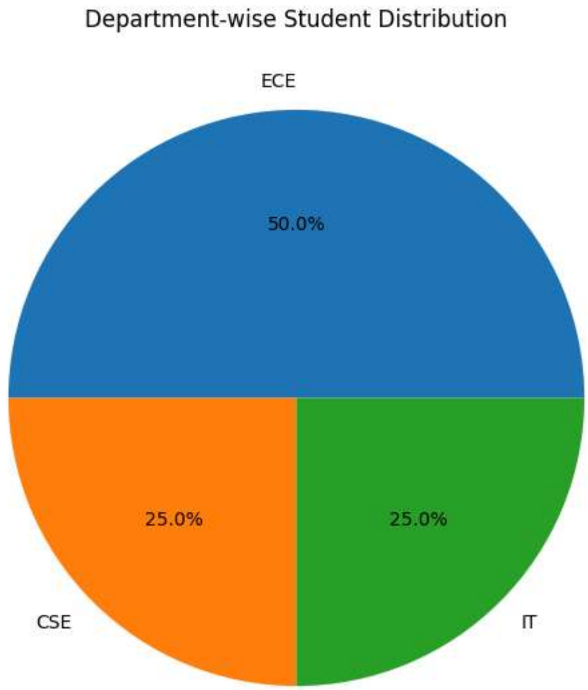
```
plot_df = df.dropna(subset=["Attendance", "Marks"])

plt.figure(figsize=(8, 5))
```

```
plt.scatter(
    plot_df["Attendance"],
    plot_df["Marks"]
)

plt.xlabel("Attendance")
plt.ylabel("Marks")
plt.title("Scatter Plot: Attendance vs Marks")

plt.show()
```



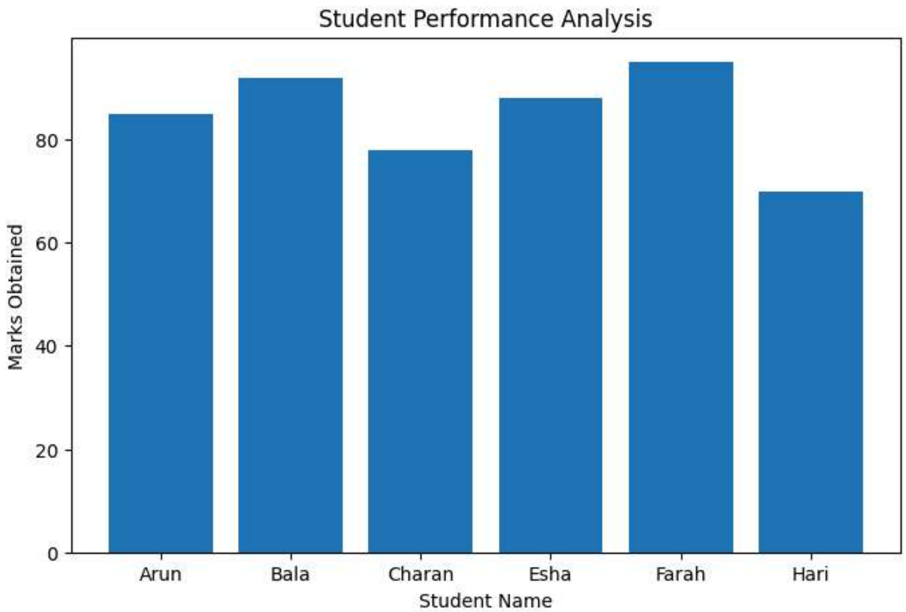
12. Pie Chart

```
department_count = df["Department"].value_counts()

plt.figure(figsize=(7, 7))

plt.pie(
    department_count,
    labels=department_count.index,
    autopct="%1.1f%%"
)

plt.title("Department-wise Student Distribution")

plt.show()
```



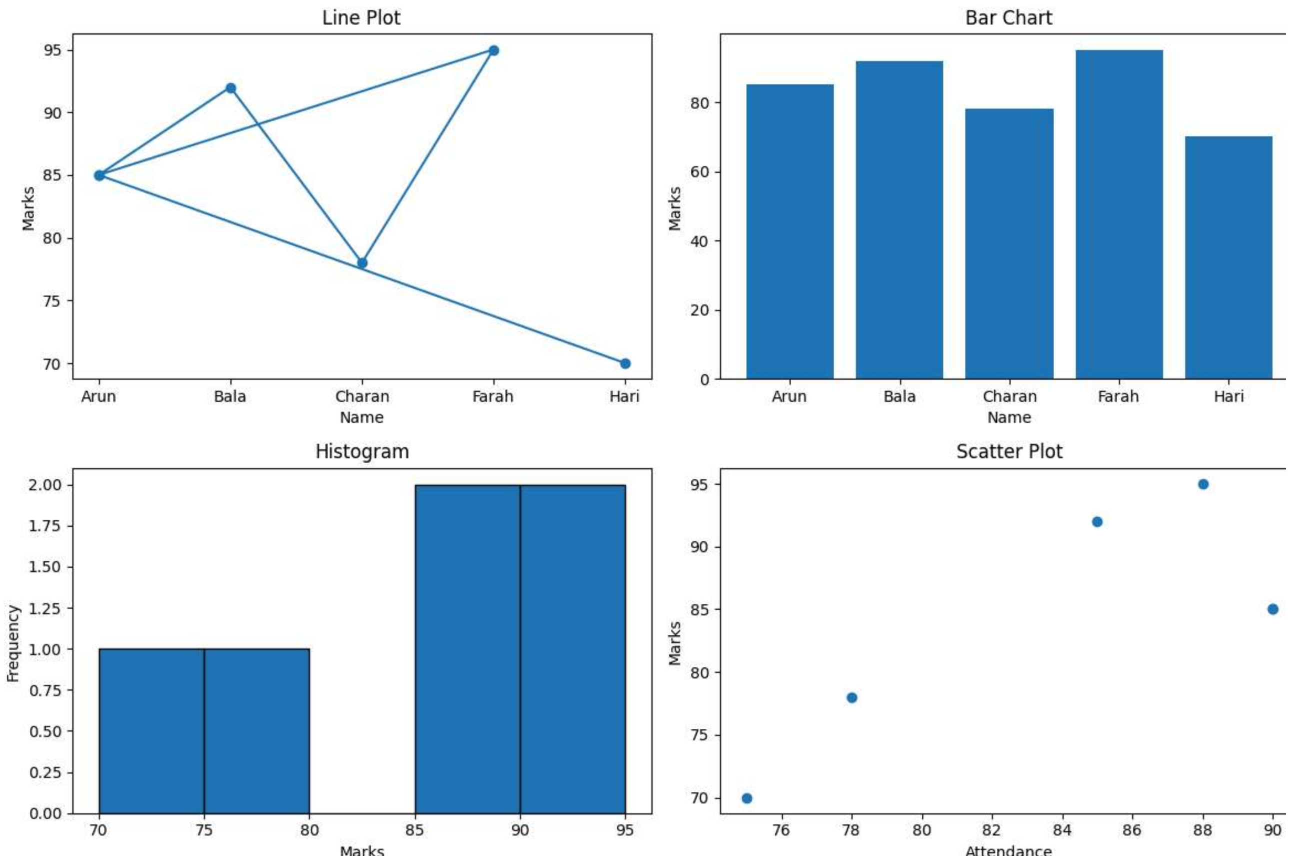
13. Plot Labels and Titles

```
plot_df = df.dropna(subset=["Marks"])

plt.figure(figsize=(8, 5))

plt.bar(
    plot_df["Name"],
    plot_df["Marks"]
)

# Plot Labels
plt.xlabel("Student Name")
plt.ylabel("Marks Obtained")

# Plot Title
plt.title("Student Performance Analysis")

plt.show()
```



14. Subplots

```
plot_df = df.dropna(subset=["Marks", "Attendance"])

plt.figure(figsize=(12, 8))

# Subplot 1 - Line Plot
plt.subplot(2, 2, 1)
plt.plot(
    plot_df["Name"],
    plot_df["Marks"],
    marker="o"
)
plt.title("Line Plot")
plt.xlabel("Name")
plt.ylabel("Marks")

# Subplot 2 - Bar Chart
plt.subplot(2, 2, 2)
plt.bar(
    plot_df["Name"],
    plot_df["Marks"]
)
plt.title("Bar Chart")
plt.xlabel("Name")
plt.ylabel("Marks")

# Subplot 3 - Histogram
plt.subplot(2, 2, 3)
plt.hist(
    plot_df["Marks"],
    bins=5,
    edgecolor="black"
)
plt.title("Histogram")
plt.xlabel("Marks")
plt.ylabel("Frequency")

# Subplot 4 - Scatter Plot
plt.subplot(2, 2, 4)
plt.scatter(
    plot_df["Attendance"],
    plot_df["Marks"]
)
plt.title("Scatter Plot")
plt.xlabel("Attendance")
plt.ylabel("Marks")

plt.tight_layout()
```


Line Plot
Marks
Name
Arun
Bala
Charan
Farah
Hari
Bar Chart
Histogram
Frequency
Scatter Plot
Attendance